

Design aspects of FMU-explore a Python module to complement PyFMI

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Jupyter notebook with Python has become a popular way to work with compiled Modelica code in the form of FMUs, see [1] and [2]. In these notebooks you can combine your own text with executable code and results in terms of diagrams and tables. The notebooks can easily be shared and even run and modified provided the FMU is also shared. It is also possible to share without any installation using Google colab [2].

The Python modules available provide flexible interaction with the FMU. The flexibility is useful for professionals who write scripts for interaction. However, the flexibility demands many details in the script that make command-line interaction cumbersome. For people with less computer background and even people with engineering background but whose focus is the results of simulations rather than developing models, can benefit from a more concise and limited direct command-line interaction. Here is a need for a simplifying layer between the user and the well-established frameworks [1] and [2].

FMU-explore is a module that addresses this simplifying needs to facilitate command-line interaction and was briefly mentioned already in [3]. Now this module is much more independent of the actual application and separated from Bioprocess Library [3] [4]. The module provides python functions and workspace dictionaries that simplify interaction and minimize the commands, parameters and diagram layouts to be entered. The functions are so far: `par()`, `init()`, `newplot()`, `simu()`, `show()`, `disp()` and `describe()` and inspired by [5].

Example 1): Generate three simulations with different initial values and show results in a pre-defined standard diagram. Note usage of Python's for-loop construct.

```
newplot()
for value in [0.7, 1.0, 1.3]: init(X_0=value); simu()
```

Example 2): Generate two simulations and in the second a parameter is changed half-way and we want to see the consequences.

```
newplot()
simu(10)
simu(4); par(a=0.4); simu(6,'cont')
```

The software has been used in both teaching and consultancy work with good results. The plan is to bring the FMU-explore module to Github during the year if there is interest.

[1] Winther C et al <https://github.com/modelon-community/PyFMI>

[2] Sommer T et al <https://github.com/CATIA-Systems/FMPy>, <https://github.com/t-sommer/fmi-cloud-tutorial>

[3] Axelsson, J. P., "Design aspects of Bioprocess Library for Modelica", OpenModelica, Linköping, Feb 2, 2021.

[4] Axelsson, J. P., "Simplified model of CHO-cultivation in Bioprocess Library for Modelica – some experience", NPCW, Copenhagen, August 22-23, 2019.

[5] Elmqvist, H., "SIMNON – User's Manual". TFRT-3091. Dept. of Automatic Control LTH, 1975.